

Lab Goal : This lab was designed to teach you more about recursion.

Lab Description : Write a program that takes in a string of unique characters and generates all possible permutations of the letters.

Sample Data :

ABC
abc
boat
it

Sample Output :

PERMUTATION OF WORD :: ABC
ABC
ACB
BAC
BCA
CAB
CBA

PERMUTATION OF WORD :: abc
abc
acb
bac
bca
cab
cba

PERMUTATION OF WORD :: boat
boat
bota
baot
bato
btoa
btao
obat
obta
oabt
oatb
otba
otab
abot
abto
aobt
aotb
atbo
atob
tboa
tbao
toba
toab
tabo
taob

PERMUTATION OF WORD :: it
it
ti

algorithm help

Make a HashSet to store the permutations of the word

Handling error scenarios

If the word is nonexistent or the empty string:

If the word is the empty string, add it to your set

Return your set

Store the first character

Store the rest of the string

Call permutations() on the remainder and store the set it gives you

Loop through each permutation in the set

Loop through each spot of the current word from rem

Insert the initial letter at the current spot

Add this permutation to your set of permutations.

Return the set of permutations

Example showing the recursion:

<u>p("CAT")</u> init = "C" words = p("AT") = {"AT", "TA"} Insert "C" at every spot in "AT" and "TA" to form "CAT", "ACT", "ATC", "CTA", "TCA", and "TAC" return answer	<u>p("AT")</u> init = "A" words = p("T") = {"T"} Insert "A" at every spot in "T" to form "AT" and "TA" return {"AT", "TA"}	<u>p("T")</u> init = "T" words = p("") = {""} Insert "T" in front of "" to form "T" return {"T"}	<u>p("")</u> return {""}
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